PRELIMINARY ANALYSIS OF WHIRLPOOL'S FREERIDING MODEL

(An anticipation of the "Bernoulli attacks")

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Motivations

Whirpool’s model is based on the idea of providing privacy by homogeneity. Whirlpool does its best to enforce this homogeneity (equal amounts inputs and outputs, uniform selection of remixed TXOs by the client and by the coordinator, etc).

Based on this principle, any heterogeneity in the structural, procedural or temporal aspects can be seen as a potential source of “leaks” that an adversary may leverage against the system.

This analysis is motivated by the observation that the number of UTXOs (re)mixed by a client remains one of the main sources of heterogeneity in Whirlpool.

The goal of this doc is to provide a first model of freeriding in Whirlpool and to investigate how the heterogenity in the number of UTXOs (re)mixed by the clients may be exploited to decrease the guarantees provided by the system.

Modelization of Whirlpool freeriding

Assumptions used for these models

A1: The number of UTXOs remixed by a targeted wallet in a specific pool is constant during the period of interest (~ target number of mixes is unbounded (infinity)).

A2: The number of UTXOs remixed by each round stays constant during the period of interest (will be set to 3 for this analysis).

A3: The number of freeriders participating to a pool is known and stays constant during the period of inteest.

A4: An unconfirmed postmix UTXO can be selected for a new mixing round.

Note: These assumptions aren’t strictly true and rely on simplifications of the real system. But most (all?) of these simplifications can be mitigated by an adversary actively monitoring the pools and the Bitcoin network.

Characterization of the selection process

Selection for a single mixing round

The selection of a postmix UTXO for a new mixing round can be modeled as 2 dependent Benouilli trials:

- The wallet (whirlpool-cli) randomly selects a postmix UTXO among u candidates (postmix UTXOs enqueued in the wallet) that will be submitted to the coordinator for the new mixing round.

- The coordinator randomly selects 3 UTXOs among the UTXOs submitted by the freeriders for the new mixig round.

The probability of a given postmix UTXO to be selected by the wallet is:

$p_w = 1 / u$ (uniform probability)

with:

u = number of UTXOs enqueud for the pool in the wallet

The probability of a submitted postmix UTXO to be selected by the coordinator is:

$p_c = 3 / r$ (uniform probability)

with:

r = number of freeriders active in the pool

Selection over a sequence of mixing rounds

The selection of a postmix UTXO over a sequence of several mixing rounds can be modeled as a Benouilli process.

Probability for an UTXO to be remixed k times over n mixing rounds

Model

By definition, the probability for a given postmix UTXO to be remixed k times during a run of n rounds can be modeled as the binomial distribution **B(n, P_w * P_c)** (conditional binomials).

$$P(k, n, P_w, P_c) = B(n, P_w * P_c) = \frac{n!}{k! * (n-k)!} * Q^k * (1-Q)^{(n-k)}$$

with:

$$Q = P_w * P_c = \frac{1}{u} * \frac{3}{r} = \frac{3}{u * r}$$

Simulations

Simulation 1: 100 freeriders / 1000 mixing rounds / 1, 10 and 100 enqueued TXOs

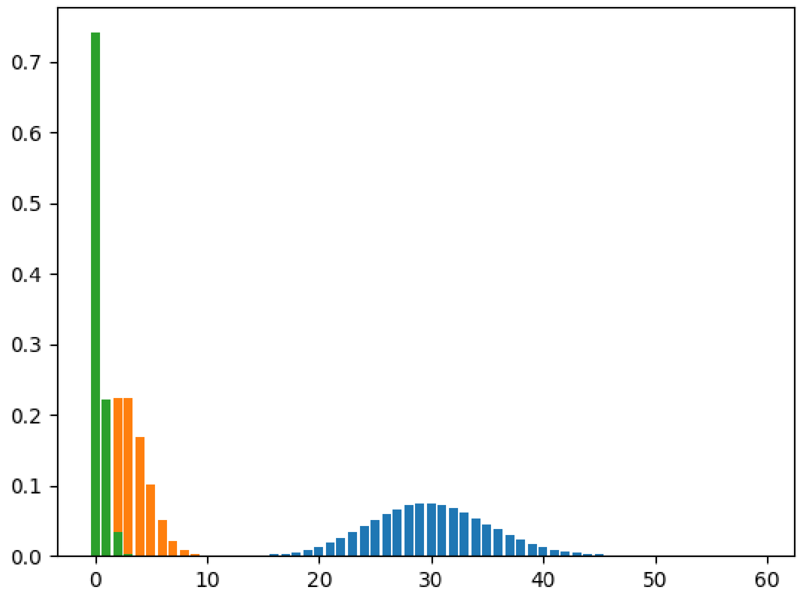


Figure 1: blue = 1 UTXO, orange = 10 UTXOs, green = 100 UTXOS

<u>**Observations**</u>

- **The lower the number of enqueued UTXOs, the higher the number of mixes expected for a given UTXO.**
- **The lower the number of enqueued UTXOs, the higher the variance (i.e. the more we can expect that remixed UTXOs will have different number of mixes).**
- The model shows that for a wallet with 100 UTXOs, it's very unlikely that its UTXOs will receive more than 2 remixes on a period of 1000 remixes. We already know that Tx0s generating a large number of UTXOs (e.g.: 100) are often problematic because of their correlation with postmix transactions consolidating a large number of UTXOs but this observation suggests that even if we exclude the case of any postmix consolidation these large Tx0s can still be targeted with a new attack described below.

Attack against large Tx0s

Rationale

Based on the default behavior of whirlpool-cli, we make the assumption that a Tx0 with a large number of premix outputs will lead to the same large number of postmix UTXOs competing for remixing.

Principle

- Starting from a large Tx0, we traverse the transactions graph and identify the descendant mix transactions.

- For each descendant transaction, we know:

    - the number of rounds processed by whirlpool between the firstmix and the descendant transaction,

    - the number of remixes on the path between the first mix and the descendant transaction.

- From these 2 values, we can compute the probability that the descendant transaction is indeed remixing one of the outputs generated by the Tx0 (~ probability that the descendant transaction is remixing one of the outputs generated by the Tx0 ~ probability that one of the outputs of the descendant transaction is related to the Tx0).

- We can also define a threshold value for this probability. Under this threshold value, we consider that the descendant output is unrelated to the Tx0.

Expected results

Since this is a probabilistic attack, false negatives are expected (for instance the firsts UTXOs being remixed will be missed) but we can expect a large number of true positives that may be enough to successfully track the postmix activity (e.g: all postmix UTXOs being individually sent to a same service).

It also means that we can probabilistically decrease the forward-looking anonset of the UTXOs generated by large Tx0s.

Generalized attack against “all” Tx0s

Rationale

Based on the default behavior of whirlpool-cli, we make the assumption that a Tx0 generating u premix outputs will lead to a minimum of u postmix UTXOs competing for remixing.

Principle

- Starting from a the Tx0, we traverse the transactions graph and identify the descendant mix transactions.

- For each descendant transaction, we know:

    - the number of rounds processed by whirlpool between the firstmix and the descendant transaction,

    - the number of remixes on the path between the first mix and the descendant transaction.

- From these 2 values, we can compute the probability that the descendant transaction is indeed remixing one of the outputs generated by the Tx0 (~ probability that the descendant transaction is remixing one of the outputs generated by the Tx0 ~ probability that one of the outputs of the descendant transaction is related to the Tx0).

- We can also define a threshold value for this probability. Under this threshold value, we consider that the descendant output is unrelated to the Tx0.

- A more conservative form of this attack might be to consider that the wallet has potentially more competing UTXOs and that only the right side of the distribution should be considered

Expected results

As stated for the attack described for large Tx0s, this attack will generate false negative due to its probabilistic nature.

It’s hard to assert what might be its impact even if it seems likely that it will be lower for wallets having a low number of competing UTXOs.

It’s also expected that the most common wallet sizes used by freeriders may lead to higher numbers of false positives.

Simulation 2: 100 freeriders / 1000, 4000, 8000 mixing rounds / 8 enqueued TXOs

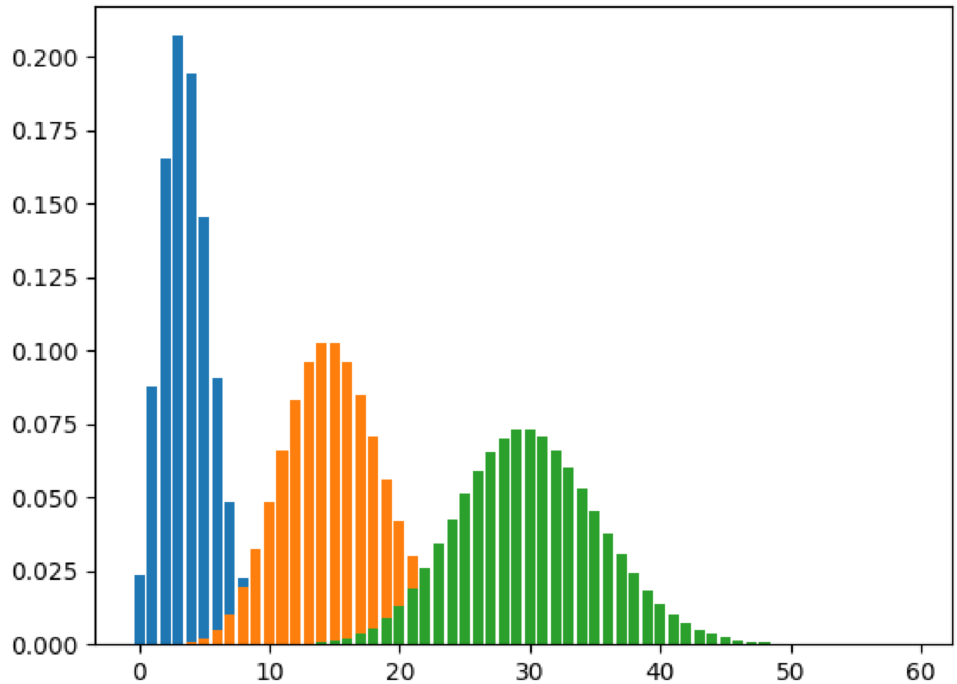


Figure 2: blue = 1000 rounds, orange = 4000 rounds, green = 8000 rounds

<u>**Observations**</u>

- **The higher the number of rounds processed by whirlpool, the higher the number of mixes expected for an UTXO**.
- **The higher the number of rounds processed by whirlpool, the higher the variance** (i.e. the more we can expect that remixed UTXOs will have different number of mixes).

Simulation 3: 100 freeriders / 1000 mixing rounds / 1, 2, 4, 8 enqueued TXOs

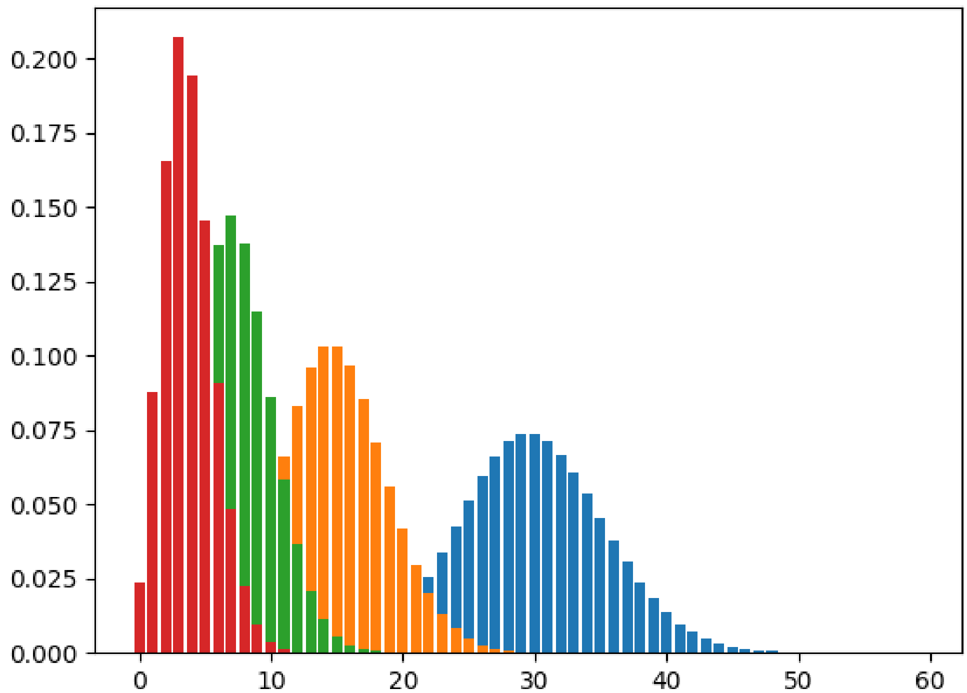


Figure 3: blue = 1 UTXO, orange = 2 UTXOs, green = 4 UTXOs, Red = 8 UTXOs

<u>**Observations**</u>

- If we compare this chart with the chart provided for Simulation 2, we get the confirmation of an intuition: We have the same probability distributions for:
    - an UTXO belonging to a wallet of 1 UTXO after 1,000 rounds processed by Whirlpool
    - an UTXO belonging to a wallet of 8 UTXOs after 8,000 rounds processed by Whirlpool

  or for:

    - an UTXO belonging to a wallet of 2 UTXOs after 1,000 rounds processed by Whirlpool
    - an UTXO belonging to a wallet of 8 UTXOs after 4,000 rounds processed by Whirlpool

More generally, if D is the probability distribution associated to a wallet W1 composed of U1 UTXOs after n rounds processed by Whirlpool, then a wallet W2 composed of U2 UTXOs (with U2 = k*U1) will have the same probability distribution D after k*n rounds processed by Whirlpool.

Simulation 4: 100 freeriders / 1000, 5000, 10000, 20000 mixing rounds / 1, 2, 4, 8 enqueued TXOs

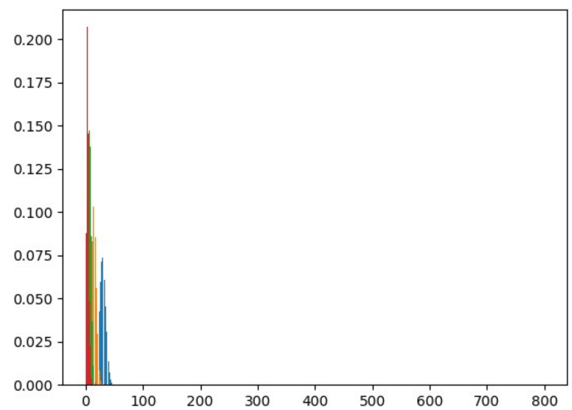


Figure 4: After 1000 rounds

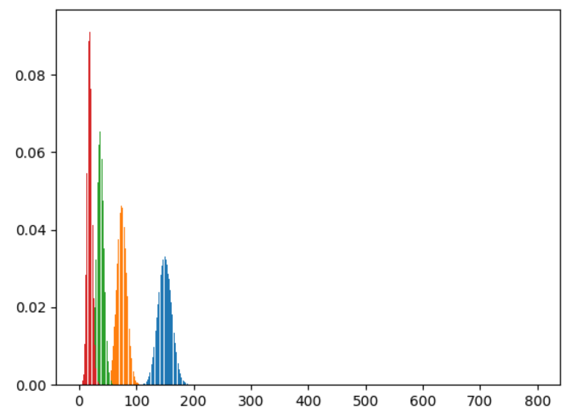


Figure 5: After 5000 rounds

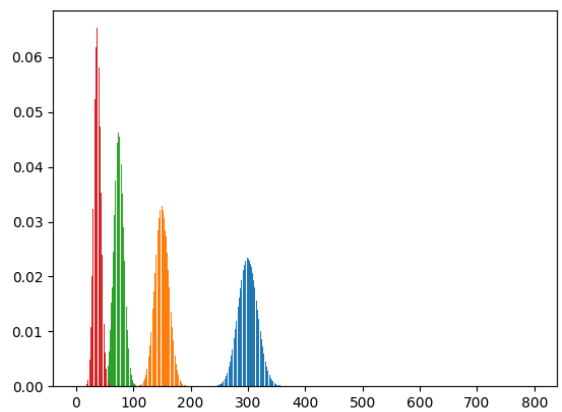


Figure 6: After 10000 rounds

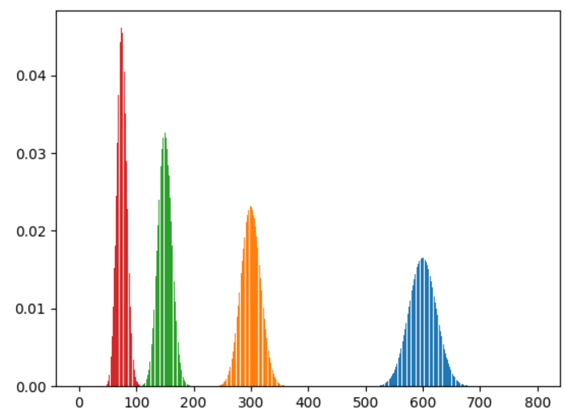


Figure 7: After 20000 rounds

Observations

- **The higher the number of rounds processed by Whirlpool, the easier the distinction between the distributions.**

Probability for an UTXO with a given age to be remixed

Model

We define the age of a postmix UTXO being remixed as:

age = j - i

with:

j = index of the mixing round spending the UTXO

i = index of the mixing round generating the UTXO

The probability for a postmix UTXO to be remixed with an age greater than or equal to a given age is:

P(age>=a, P_w, P_c) = (1 – P_w * P_c) ^ (a-1) = (1 – 3 / (u * r)) ^ (a-1)

(i.e. probability that UTXO isn’t selected for a run of a-1 remix rounds processed by Whirlpool)

The probability for a postmix UTXO to be remixed with an age lower than or equal to a given age is:

P(age<=a, P_w, P_c) = 1 - P(age>=a, P_w, P_c) = 1 - (1 – 3 / (u * r)) ^ (a-1)

Simulation 1: 100 freeriders / 1, 2, 4, 8 and 16 enqueued TXOs

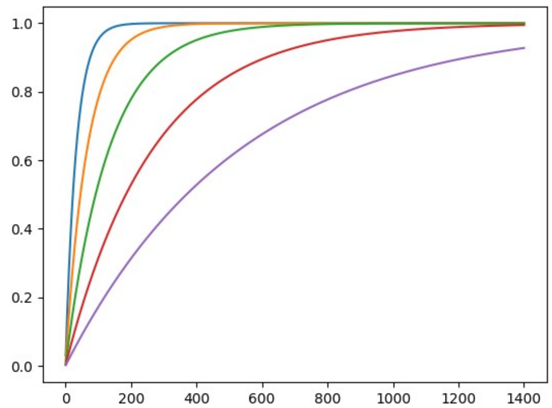


Figure 8: P(age <= x) blue = 1 UTXO, orange = 2 UTXOs, green = 4 UTXOs, red = 8 UTXOs, purple = 16 UTXOs

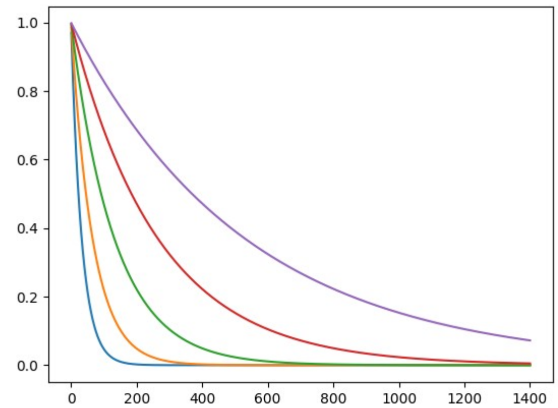


Figure 9: P(age >= x) blue = 1 UTXO, orange = 2 UTXOs, green = 4 UTXOs, red = 8 UTXOs, purple = 16 UTXOs

<u>Observations</u>

- The model shows that for a wallet with u UTXOs, it’s unlikely that its UTXOs will be remixed with an age higher than a given value (depending on u). This observation suggests that we can define a new attack targeting an isolated remix (instead of a chain of remixes). For instance, an UTXO controlled by a wallet remixing a single UTXO has a probability of 0.01% to be remixed with an age greater than 300 rounds. Thus, we may consider that the remix of a 300 rounds old UTXO is very unlikely to be related to a wallet remixing a single UTXO.

Attack against a single mix

Rationale

The goal of this attack is to analyze an individual remix and to estimate the probability that this remix is unrelated to a targeted wallet remixing a given number of UTXOs.

Principle

- Define a threshold value t for the age probability. Under this threshold value, we will consider that the remix is unrelated to the targeted wallet.

- Compute P(age >= a) and P(age <= a) with u being the estimated number of UTXOs remixed by the targeted wallet.

- If u is small and if P(age >= a) <= t we consider that the remix is unlikely to be related to the targeted wallet.

- If u is large and if P(age <= a) <= t we consider that the remix is unlikely to be related to the targeted wallet.

Expected results

Unlike the first attacks described in this document, the results provided by this attack will be sensitive to the accuracy of the estimated number of UTXOs remixed by the wallet. The more this figure is underestimated for small wallets, the more false positives produced by the attack.

Preliminary discussion

This modelization of Whirlpool's freeriding confirms the intuition that a lower number of UTXOs enqueued in the wallet leads to a higher expected remixing rate for each individual UTXO.

It also higlights a few less intuitive characteristics (at least for me) of the remixing process:

- absent a limit capping the number of remixes, a wallet with a lower number of enqueued UTXOs will spread the UTXOs over a larger range of number of remixes for a given number of rounds processed by Whirlpool.
- absent a limit capping the number of remixes, remixing the UTXOs enqueued in a wallet on a larger period of time will spread the UTXOs over a larger range of number of remixes.

It also highlights that Tx0s generating a large number of premix UTXOs are more prone to targeted attacks, even if we exclude the case of postmix transactions consolidating a large number of postmix UTXOs.

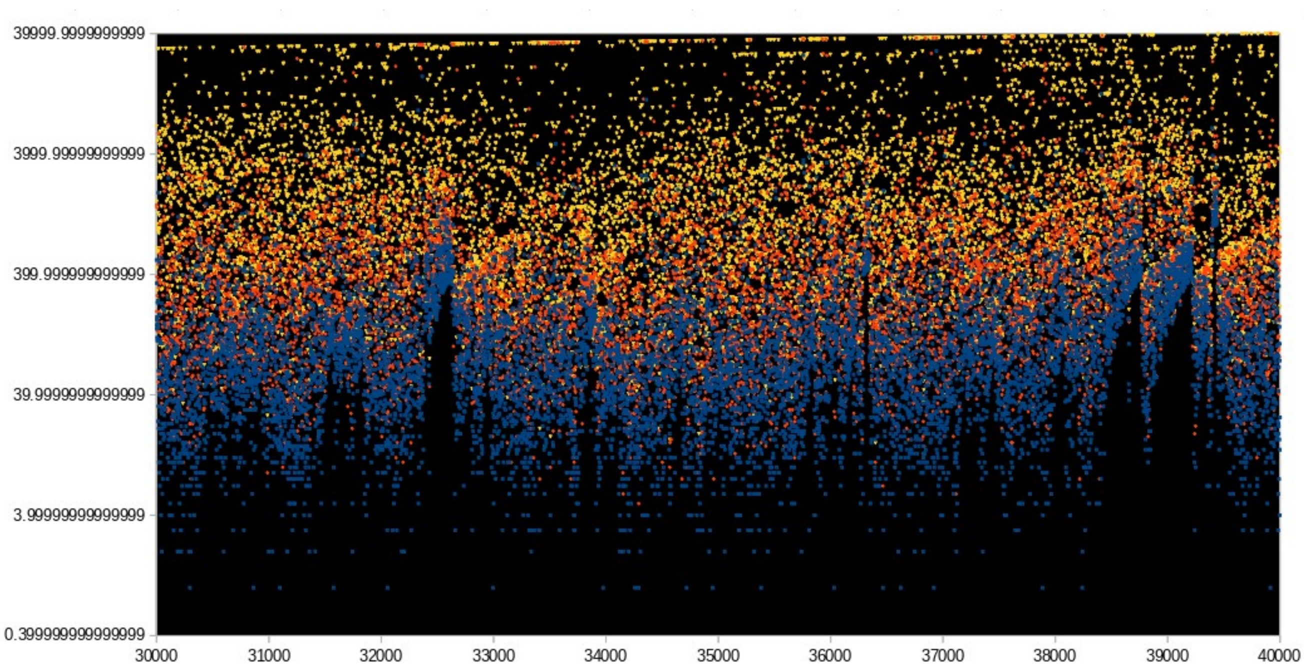
In general, this modelization suggests that the more the number of UTXOs enqueued by the clients is homogeneous (spread over a small range), the better. The theoretically perfect (but practically infeasible) scenario being all clients having the same number of enqueued UTXOs.

While the model suggests that limiting the number of outputs generated by a Tx0 should be a net benefit, it also shows that this solution won't be miraculous. Over a long enough period of time, even a small difference in the number of enqueued UTXOs will lead to a large divergence of the distributions between the UTXOs of 2 wallets.

An open question that certainly deserves to be investigated is if the use of a different distribution for the selection processes (instead of the uniform distribution currently used) helps to mitigate the issue.

Annexes

Age of remixed TXOs for rounds 30k to 40k (pool 0.01BTC)



Distribution of remixed TXOs by age for rounds 30k to 40k (pool 0.01BTC)

TL/DR: Mode ~= 25, Median ~= 290

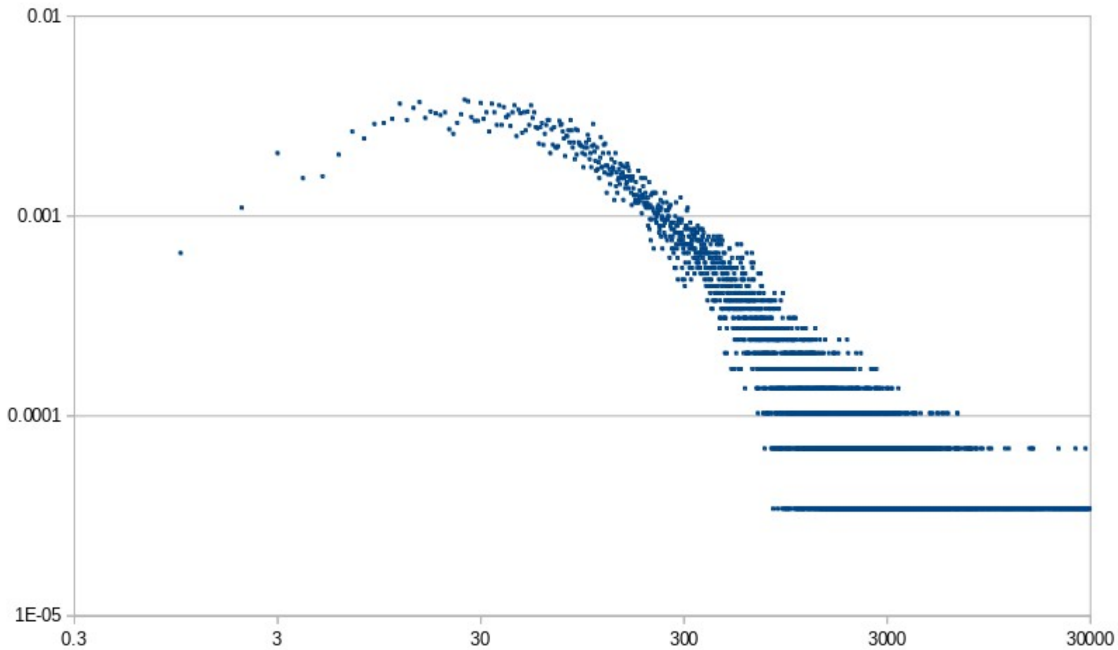


Figure 10: Distribution of remixed UTXOs by age (loglog scale)

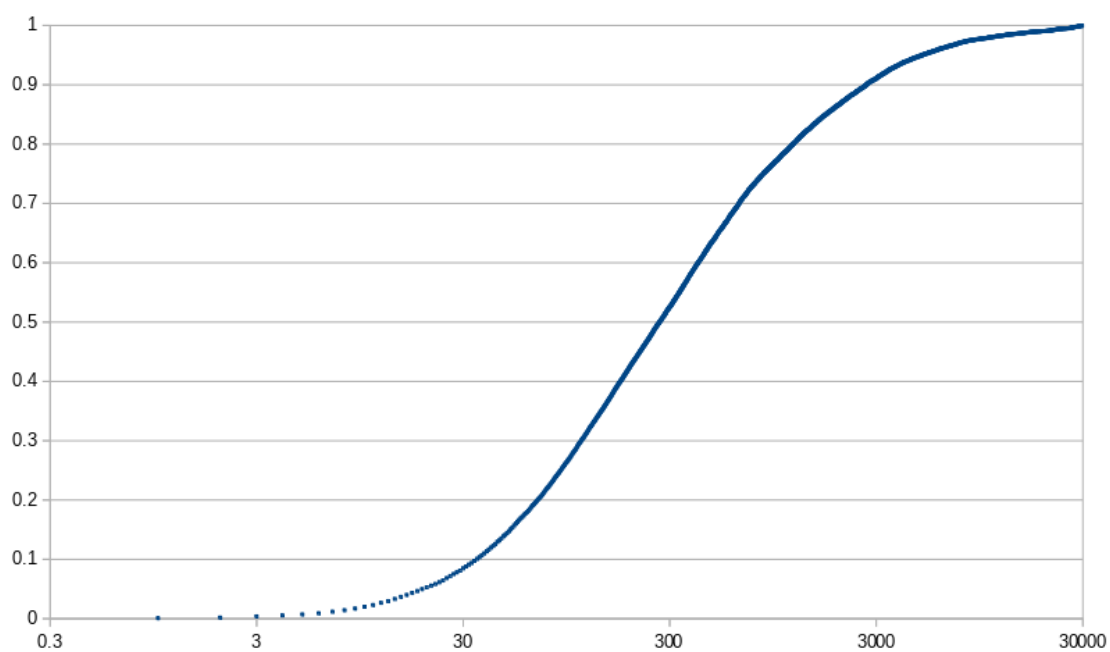


Figure 11: Cumulated distribution of remixed UTXOs by age (loglin scale)